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Journal of King Saud University – Computer and Information Sciences

journal homepage: www.sciencedirect.com

A three-stage layer-based heuristic to solve the 3D bin-packing problem under balancing constraint

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ARTICLE INFO

Article history:

Received 26 April 2021

Revised 30 June 2021

Accepted 8 July 2021

Available online 15 July 2021

Keywords:

3D bin-packing problem

Layers

Load balancing

Box orientation

ABSTRACT

This research investigates the Three Dimensional-Single Bin-Size Bin-Packing Problem (3D-SBSP). The aim is to pack n boxes of different dimensions and weights into a minimum number of equally balanced bins. The boxes can be rotated according to six possible orientations. Balancing the bins, volume and weight wise, is of high importance. To achieve this goal, a three-stage heuristic was developed. In the first stage, the different possible layers are generated. Each layer is formed by boxes of same type packed according to a common given orientation. The second stage generates solution candidates using the layers formed in stage 1. In stage 3, the boxes to be shipped are placed in the bins following the solution candidates of stage 2 that give the best volumetric and/or weight utilization. The proposed heuristic was successfully validated through an exhaustive experimental study using a variety of problem instances and involving comparisons with other existing heuristics.

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1. Introduction

The bin-packing problem is a strongly NP-hard combinatorial optimization problem, Korte et al. (2012). The problem is to find a suitable packing of items with different sizes and weights into a minimum number of bins of equal or different capacities. There are many variations of this problem depending on the bins' dimensions; whether identical or different dimensions and how the items are packed; one-dimensional, two-dimensional, or three-dimensional, Wäscher et al. (2007). In real world cases, the aim is not only to determine a packing with fewest bins but also to generate balanced bins. In fact, the center of mass of loaded bins into an aircraft cargo, trucks, or container ships should fall as close as possible to an ideal specified point. Balancing the bins leads to an increase of the safety and the efficiency of the transport as well as minimizes the fuel consumption, Mongeau and Bes (2003). Balancing the bins while loading the items with the aim of using the minimum number of bins, is a multiple conflicting objectives problem.

This research was motivated by the problem's practical applications in various fields such as industrial courier, container ship loading, and production industry.

This research investigates the Three Dimensional-Single Bin-Size Bin-Packing Problem (3D-SBSP). The focus is to determine the best orthogonal placement of the items in the minimum number of bins. The items have to be placed without overlapping according to a flexible rotation. We present a three-stage layer-based heuristic that generates feasible solutions showing which items to be placed in each bin and the way they are arranged. The first stage forms different layers each containing identical items according to a particular orientation. During the second stage, feasible loading space utilization solutions are generated using the best possible combination of different layers developed in stage 1. Finally, a feasible solution (in term of bin's volume and weight utilization) with a minimum number of balanced bins is generated.

This paper is organized as follows. In Section 2 related literature about the 3D bin-packing problem is analyzed. In Section 3 a detailed description of the problem's constraints and objectives is presented as well as the proposed three-stage heuristic to solve the problem. Section 4 reports the computational results. Finally, Section 5 concludes the paper and provides some future works.

Peer review under responsibility of King Saud University.



Production and hosting by Elsevier

E-mail address: yharrath@uob.edu.bh<https://doi.org/10.1016/j.jksuci.2021.07.007>

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2. Literature review

The classical multidimensional bin-packing problem considers the minimization of number of bins. This problem was widely studied in the literature. Exact algorithms, heuristics and lower bounds have been developed. In Martello and Vigo (1998), the authors have studied the two-dimensional bin-packing problem and proposed the first exact method based on branch and bound algorithm. The authors in Martello et al. (2000) investigated the 3D case and introduced the discrete set of Corner Points to reduce the search space of their previous exact method. In Fekete and Schepers (2004) and Fekete et al. (2007) an implicit representation of multidimensional packings by means of interval graphs have been defined. The authors consider the relative position of the boxes in a feasible packing and then define a graph describing the overlapping of the items in the bins based on their projection on each orthogonal axis. Following such characterization, the authors proposed a two-level search tree. An algorithm using the concept of block composed of one or two types of boxes was developed in Eley (2002). In addition, the authors in Zhang et al. (2012) have proposed a block-loading algorithm based on multi-layer search for the single bin 3D packing problem.

A variant of the studied problem focuses on balancing the loads of the used bins. Recently, this problem attracted some researchers although such additional constraints increase its complexity. Mathematical formulations to solve the problem of determining the arrangement of a set of items in a cargo that optimizes the load balance in a single bin have been proposed in Kaluzny and Shaw (2009). The same problem was investigated in Junqueira et al. (2012) with the consideration of more realistic constraints such as the vertical and horizontal stability and the load bearing strength of the cargo. To solve this problem, the authors proposed mixed integer linear programming models. In addition to the minimization of the unused volume in the loaded bins, many additional constraints encountered in real world applications such as the stability, the fragility of the items, the weight distribution, and the possibility to rotate the boxes, Paquay et al. (2016). A mixed integer linear program was proposed and validated on small instances. The authors in Trivella and Pisinger (2016) focused on finding a packing of the items requiring the minimum number of balanced bins. The load balancing is handled by forcing the bin's center of mass to lie inside a stable region. A heuristic algorithm that sorts the items according to a descending order of their weights and simultaneously places them in a predefined minimum number of bins was proposed in Kaabi et al. (2018) where additional bins are used when needed.

Constructive layer-based approaches have been developed in the literature to ensure the load balancing. A real world single truck packing problem under load stability and box orientation constraints, weight limit of the truck, and uniform distribution of weights inside the truck was addressed in Costa and Captivo (2016). A constructive heuristic based on a layer arrangement approach that comprises a corner selection strategy for placing the boxes was proposed. In Saraiva et al. (2015), a layer-building algorithm for the multiple bin-packing problem was proposed. The algorithm first builds horizontal layers of identical items and then greedily generates packing patterns by loading one layer at a time. Similarly, the authors of Ebrahim et al. (2019) developed a layer-based approach comprised of three stages to obtain balanced filled bins. Layers in a bin are formed by identical items of similar orientation.

The rotation of the items is an additional constraint that is considered in limited number of researches. The problem where the rotation is allowed is called non-oriented bin-packing problem. In Dell'Amico et al. (2002), the first lower bound for the

non-oriented model of the 2D bin-packing problem was proposed. Different scenarios of the same problem were also studied by Clautiaux et al. (2007). While in Boschetti (2004), the authors investigated the 3D version of the problem and developed 2 new lower bounds. A more recent related work has been published in Liao and Hsu (2013) where new better theoretical lower bounds were proposed.

3. Solution approach

3.1. Assumptions and constraints

The studied 3D bin-packing problem consists in loading n cuboid boxes into a set of identical cuboid bins of equal capacity weight wise and volume wise. Boxes are classified into different types according to their dimensions. We considered the non-oriented model where boxes are allowed to be rotated according to six different orientations as illustrated in Fig. 1. The constraints for this problem are as follows.

- Each box should be entirely placed inside the bin.
- Boxes cannot overlap within the bin.
- Each packed box is parallel to the bin's surfaces.
- The total weight of the loaded boxes in a given bin should not exceed its maximum allowed weight.

The objective is to pack the boxes into a minimum number of balanced bins. Following are the main notations used to describe the problem.

- t : number of box types: dimension wise
- n_i : number of boxes of type $i, i = 1 \dots t$
- n : the total number of boxes of all types: $n = \sum_{i=1}^t n_i$
- m : the number of bins used to pack the n boxes
- D, L, H : depth, length and height of the bin
- W : maximum allowed weight of the bin
- d_i, l_i and h_i : depth, length and height of a box of type $i, i = 1 \dots t$
- w_k : weight of a box $k, k = 1 \dots n$
- w^i : maximum allowed weight of a box of type $i, i = 1 \dots t$
- S_1 : side of the bin that is parallel to x-axis of length L
- S_2 : side of the bin that is parallel to y-axis of length H
- S_3 : side of the bin that is parallel to z-axis of length D

3.2. Three-Stage Layer-Based Heuristic (TSLBH)

3.2.1. Stage 1

In the first stage, all feasible horizontal layers of identical boxes are generated by considering the six boxes' orientations. To form a horizontal layer, the maximum number of identical boxes are placed parallel to the base of the bin according to one of their six orientations illustrated in Fig. 1. Any layer will contain boxes of same type all placed in a given orientation. In total, $6t$ different horizontal layers can be formed. The steps to form horizontal layers are shown in Algorithm 1. It is trivial that this algorithm has a linear complexity $O(t)$. In addition, in practice t is not a large number since the number of different box types is limited as was confirmed by the collected data from FedEx.

Algorithm 1. Stage 1: Generating horizontal layers

1:in: $t, D, L, d_i, l_i, h_i, i = 1, \dots, t$ out: n_{ip} ,
 $(i = 1, \dots, t, p = 1, \dots, 6)$ (maximum number of boxes per one horizontal layer for each box type i and each box orientation p .)

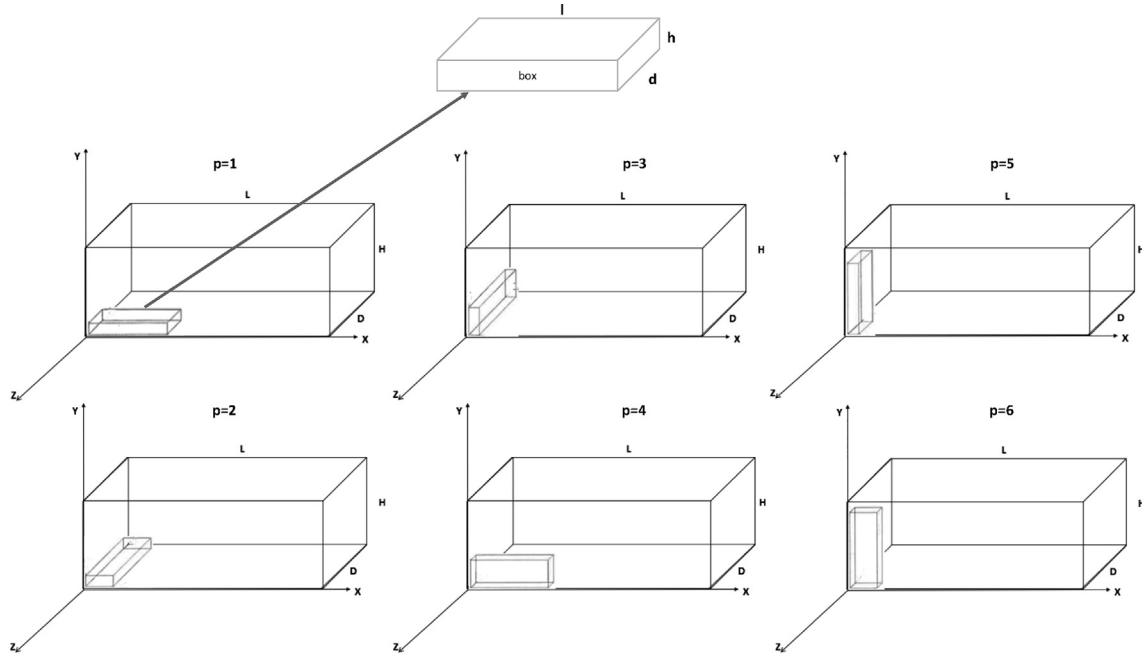


Fig. 1. The six possible box orientations.

```

2: for  $i \leftarrow 1$  to  $t$  do
3:   for  $p \leftarrow 1$  to 6 do
4:      $n_{i,p} \leftarrow \lfloor \frac{D}{d_{i,p}} \rfloor \times \lfloor \frac{L}{l_{i,p}} \rfloor$ 
5:   return  $n_{i,p}$ 
6: end for
7: end for

```

In step 4 of Algorithm 1, $d_{i,p}$ represents the length of the side of box type i parallel to S_3 according to orientation p and $l_{i,p}$ is the length of the side of box type i parallel to S_1 according to orientation p . For example, $d_{1,3} = l$ and $l_{1,3} = h$.

3.2.2. Stage 2

In this stage, all feasible solutions with respect to the bin's volume are generated. This is achieved through filling a bin using different combinations of layers found in stage 1. A feasible solution can be formed by vertically placing horizontal layers, each containing boxes of the same type. The selected layers in a bin can differ in boxes types or boxes orientations. Algorithm 2 details the process of generating the solution candidates. Note that $h_{i,p}$ represents the length of the side of box type i parallel to S_2 (vertical side) according to orientation p .

Algorithm 2. Stage 2: Generation of feasible solutions with respect to the bin's volume.

```

1: in:  $t, D, L, H, d_i, l_i, h_i, i = 1, \dots, t$  out:  $S$ : Set of feasible solutions.
2:  $S \leftarrow \emptyset$ 
3: for  $p \leftarrow 1$  to 6 do
4:   for  $r_{1,p} \leftarrow 1$  to  $\lfloor \frac{H}{h_{1,p}} \rfloor$  do
5:     for  $r_{2,p} \leftarrow 1$  to  $\lfloor \frac{H}{h_{2,p}} \rfloor$  do
6:       .
7:       .
8:       .
9:     for  $r_{t,p} \leftarrow 1$  to  $\lfloor \frac{H}{h_{t,p}} \rfloor$  do

```

```

10:   if  $(\sum_{i=1}^t r_{i,p} \times h_{i,p} < H)$  then
11:      $S_{i,p} \leftarrow \bigcup_{i=1}^t r_{i,p}$  layers of boxes type  $i$  according to orientation  $p$ 
12:      $S \leftarrow S \cup S_{i,p}$ 
13:   end if
14: end for
15: end for
16: end for
17: end for
18: return  $S$ 

```

The variable $r_{i,p}$ $i = 1, \dots, t$ $p = 1, \dots, 6$ used in Algorithm 2 is the number of layers containing boxes of type i according to orientation p that can vertically fit in a bin. As there number of different box types in practice is limited due to arrangement purposes, therefore the number of nested loops in this algorithm is not huge. In addition, the upper limits of the number of iterations of the inner loops given by the formula $\lfloor \frac{H}{h_{i,p}} \rfloor$ is a small constant resulting from the division of a bin's dimension by a dimension of a box type i . As a result, Algorithm 2 has a constant complexity and can generate the set of feasible solutions S in a very short time. Note that Algorithm 2 generates feasible solutions that only respect the bin's volume constraint. The bin's weight constraint will be taken into account in Algorithm 3 of stage 3.

3.2.3. Stage 3

Stage 3 of the proposed approach is inspired from the LBA-3DBPP algorithm developed by Ebrahim et al. (2019). This stage allows with the help of the outcome of Algorithm 2 to generate a feasible solution by loading the boxes into a minimum number of bins. The pseudocode shown in Algorithm 3 illustrates the steps of this stage. The main idea is to iteratively fill a bin by checking the number of unpacked boxes of each type. At each iteration we select the solution candidate generated in Algorithm 2 that uses the maximum of the bin's volume and respects its maximum allowed weight. If the total volume of the remaining unpacked

boxes is smaller to the volume of a bin, then the last bin is filled and the algorithm stops.

Note that the condition of the while loop in line 6 ensures that there are enough boxes of each type to form a solution. Otherwise a new bin is used to place the remaining boxes even though the total weight of the last bin might have a negative effect on the load balance objective.

The nested for loops between lines 11 and 14 fills in a sequential manner the empty bins using the best possible solution candidates from the set S . Each bin is filled in a way that it contains a mixture between heavy and light boxes to guarantee certain degree of balance between the bins. The for loop between lines 15 and 19 updates the set of remaining non-packed boxes by removing those added to the current bin and moves to filling the next bin.

Algorithm 3 has a polynomial complexity as shown in Lemma 1. The term $(\frac{H}{\min(h_{i,p})})^t$ does not have a negative impact on the algorithm complexity as the fraction is a constant and t has in general very small values.

Lemma 1. Algorithm 3 has a complexity $O(\max(tn^2 \log n, nt(\frac{H}{\min(h_{i,p})})^t))$

Proof 1. Algorithm 3 includes a while loop in line 6 which iterates n times. The for loop from line 7 to line 9 turns t times in each of them it sorts the remaining n_i items. As the sorting task can be done in approximately $n \log n$ time, therefore, the time complexity of the algorithm from line 6 to line 9 belongs to $O(tn^2 \log n)$. The while loop in line 6 joined with the nested for loops from line 11 to line 22 have a complexity $O(nt|S|)$. According to Algorithm 2, the size of the set S can be increased by 1 in the innermost loop. As there are t nested loops in that algorithm each turning $\lfloor \frac{H}{h_{i,p}} \rfloor$ iterations, therefore the size of the set S can be bounded as shown in Eq. (1).

$$|S| \leq \prod_{i=1}^t \lfloor \frac{H}{h_{i,p}} \rfloor \leq (\frac{H}{\min(h_{i,p})})^t \quad (1)$$

Algorithm 3. Stage 3: Generation of a feasible solution (balanced bins)

```

1: in:  $t, W, n_i \ i = 1, \dots, t, n$  out:  $m$  filled balanced bins.
2:  $S \leftarrow \{S_1, S_2, \dots, S_r, \dots, S_{|S|}\}$  set of solution candidates generated in stage 2
3:  $n_{ir} \leftarrow$  number of boxes of type  $i$  in solution  $S_r$ ,  $i = 1, 2, \dots, t; r = 1, 2, \dots, |S|$ 
4:  $m \leftarrow 0$ 
5: Sort  $S$  into a decreasing order of bin's volumetric utilization
6: while (There are boxes to be packed) do
7:   for  $i \leftarrow 1$  to  $t$  do
8:     Sort the remaining  $n_i$  unpacked boxes into a decreasing order of their weights
9:   end for
10:   $W_{S_r} \leftarrow 0$ 
11:  for  $r \leftarrow 1$  to  $|S|$  do
12:    for  $i \leftarrow 1$  to  $t$  do
13:       $W_{S_r} \leftarrow W_{S_r} + \text{weight of the } n_{ir}/2 \text{ heaviest boxes} + \text{weight of the } n_{ir}/2 \text{ lightest boxes}$ 
14:    end for
15:    if ( $W_{S_r} < W$ ) then
16:       $m \leftarrow m + 1$ 

```

```

17:   for  $i \leftarrow 1$  to  $t$  do
18:      $n_i \leftarrow n_i - n_{ir}$ 
19:   end for
20:   break
21: end if
22: end for
23: end while

```

4. Computational experiments

In order to test and validate the proposed heuristic, an extensive experimental study was conducted using real parameters from FedEx. In fact, the bin dimensions and its maximum weight capacity, the box types and their possible dimensions and weights as well as the range of number of boxes that can be received each day are all taken from a local FedEx office. One type of bin and three types of boxes were considered. The dimensions of the bin as well as the boxes are displayed in Table 1. Unfortunately, we couldn't get real data for the number of boxes and their weights, that is why we used a random computational generation of twenty different problem instances with number of boxes n in the range $[100, 200000]$. The weight $w_k, k = 1 \dots n$ of each box of type $i, i = 1 \dots t$, is randomly set in the interval $[0, w^i]$. The three stages of the heuristic were implemented using Visual Studio 2017 C# language and MS SQL Server 2014 database to store the bin and boxes parameters and shipments details. A 64-bit Windows 10 Lenovo PC, with 8.0 GB DDR4 SDRAM, IntelCore™ i7 – 6700CPU@3.40GHz, 4 cores and 8 logical processors was used.

4.1. Results of stage 1

The results of implementing stage 1 of the proposed heuristic are summarized in Table 2. Every value $n_{i,p}$ in that table represents the maximum number of boxes of type i forming a layer according to a given box orientation p . The advantage of stage 2 is to provide diverse options for loading the boxes in the bins according to different orientations. Therefore, a maximum volumetric use of bins will be guaranteed.

4.2. Results of stage 2

Two experiments were conducted with 2 and 3 box types respectively. In experiment one, where only boxes of type 1 and 2 are considered, 3183 solution candidates have been generated. The highest bin's utilization percentage reached the value of 93.6 whereas the lowest bin's percentage utilization is 12.1. These results are better than those shown in Ebrahim et al. (2019) where only 16 solution candidates have been generated with bin's utilization percentages between 91.6 and 17.8.

In experiment two, where three box types are considered, 59150 solution candidates have been generated. The highest bin's utilization percentage reached the value of 96.3 whereas the lowest bin's percentage utilization is 10. Again, these results are better than those of Ebrahim et al. (2019) where only 69 solution candidates have been generated with bin's utilization percentages between 94.4 and 10.2.

4.3. Results of stage 3

As mentioned earlier, in stage 3 the boxes are loaded in a minimum number of balanced bins with respect to their maximum

Table 1

Bin and boxes dimensions (in cm and kg).

	Depth	Height	Length	Max. weight
Bin	243.8	178	317.5	6804
Box type 1	40.16	32.86	25.88	10
Box type 2	54.8	42.1	33.5	25
Box type 3	22.2	19.7	28.7	13

Table 2

Maximum number of boxes in each layer generated in stage 1.

p	$n_{1,p}$	$n_{2,p}$	$n_{3,p}$
1	49	35	112
2	54	36	110
3	81	45	132
4	84	49	128
5	63	25	168
6	72	28	160

allowed weight. The best feasible solution candidates generated in stage 2 are selected to form the final solution. It's worth mentioning that the weight constraint on the solution candidates generated

in stage 2 is satisfied in experiment 1 but not always in experiment 2 due to the boxes' parameters. Table 3 summarizes the results obtained by the proposed heuristic TSLBH where box types 1 and 2 are used. In term of number of used bins, it is clear that TSLBH outperforms the two heuristics proposed in Kaabi et al. (2018) and Ebrahim et al. (2019) in 18 out of the 20 tested instances. While TSLBH generates feasible solutions with same number of bins for the other two instances. We noticed that TSLBH fairly distributes the boxes among the used bins in term of load balance and volume utilization percentages in all tested instances. Fig. 2 shows the results for $n = 3000$ boxes of types 1 and 2. The 15 filled bins generated by TSLBH have volumetric utilization between 82% and 94% and total weights between 27% and 40% of the bins' maximum capacity. The bins filled by TSLBH are more balanced than those filled by LBA-3DBPP volume and weight wise.

The above observations are also confirmed through experiment 2 for three box types as shown in Table 4 and Figs. 3–5 for the two problem instances with $n = 3000$ and $n = 10000$ boxes respectively.

It was noticed that TSLBH provides feasible solutions in a very short time even for large-sized problem instances which makes TSLBH suitable to solve real time bin packing problems.

Table 3

Experiment 1: Computational results for box types 1 and 2.

n	Kaabi et al. (2018)		LBA-3DBPP		TSLBH	
	bins	time (s)	bins	time (s)	bins	time (s)
100	1	< 1	1	< 1	1	< 1
500	3	< 1	3	< 1	3	< 1
1000	6	< 1	6	< 1	5	< 1
2000	13	< 1	11	< 1	10	< 1
2500	15	< 1	13	< 1	13	< 1
3000	19	< 1	16	< 1	15	< 1
4000	25	< 1	21	< 1	20	< 1
5000	30	< 1	27	< 1	25	< 1
6000	37	< 1	32	1.7	30	1.6
7000	43	< 1	37	1.3	35	1.5
7500	46	< 1	40	1.4	38	1.6
8000	50	< 1	42	1.6	40	1.8
9000	56	< 1	47	1.8	45	2.3
10000	60	< 1	53	2.2	50	2.3
25000	153	1.7	131	6.8	124	7.4
50000	305	6.9	261	17.3	248	22.3
75000	457	20.3	392	29	373	27.7
100000	609	51.5	522	48.4	497	42
150000	913	164.3	783	97.7	745	77.2
200000	1217	329.8	1044	166	993	131.1

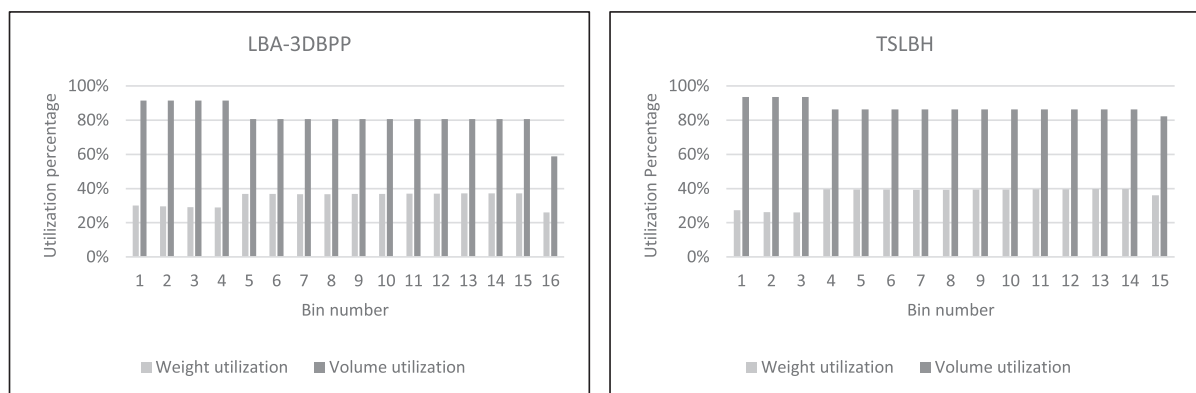
**Fig. 2.** Load balancing and volume utilization percentages: LBA-3DBPP vs TSLBH ($n = 3000$, 2 box types).

Table 4

Experiment 2: Computational results for three box types.

<i>n</i>	LBA-3DBPP		TSLBH	
	bins	time (s)	bins	time (s)
100	1	< 1	1	< 1
500	3	< 1	2	1.3
1000	5	< 1	5	2
2000	9	< 1	9	3.6
2500	12	< 1	11	4.7
3000	14	< 1	13	5.3
4000	18	< 1	18	7.6
5000	23	< 1	22	8.9
6000	27	1.1	26	10.9
7000	32	1.3	30	13.1
7500	34	1.4	33	14.6
8000	36	1.6	35	14.8
9000	41	1.9	39	17.2
10000	45	2.3	43	19.3
25000	113	7.5	107	46.4
50000	225	23.5	214	99
75000	337	30.7	322	153.4
100000	450	45.1	429	209.3
150000	674	86.5	643	334.3
200000	899	149.1	857	465.6

5. Conclusion

This paper studied a non-oriented 3D bin-packing problem. Two objectives have been considered namely the minimization of number of bins used to pack different types of boxes and bins' load balance. The load balance objective aims to equally distribute the load among the bins weight and volume wise. A 3-stage heuristic TSLBH was proposed to solve the given problem. The main idea of TSLBH is to fill the bins layer by layer according to different box orientations and optimizing the above mentioned objectives. TSLBH was implemented and thoroughly tested with problem instances of different sizes. The computational results showed that TSLBH outperforms other listed heuristics for both objectives as well as the execution time.

Further studies can be conducted to improve TSLBH by considering more types of boxes. Furthermore, a bin's layer can be formed using different types of boxes placed in different orientations. Other constraints might be imposed on the box placement such as box content (liquid, fragile, etc.), bins' dimensions etc.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

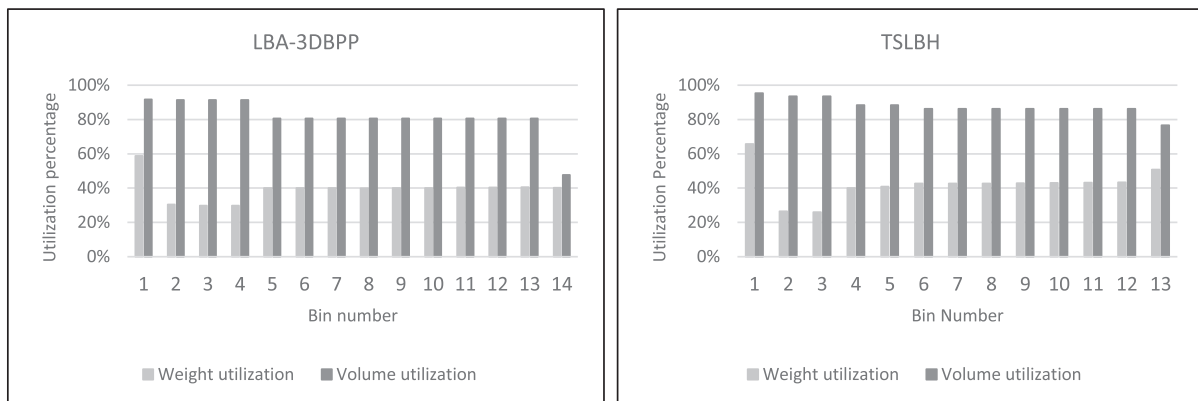


Fig. 3. Load balancing and volume utilization percentages: LBA-3DBPP vs TSLBH ($n = 3000$, 3 box types).

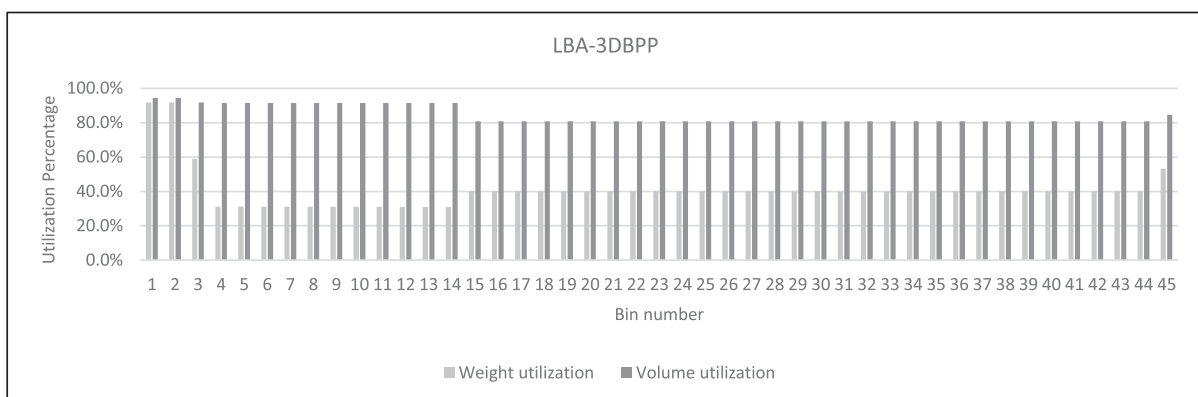


Fig. 4. Load balancing and volume utilization percentages: LBA-3DBPP ($n = 10000$, 3 box types).

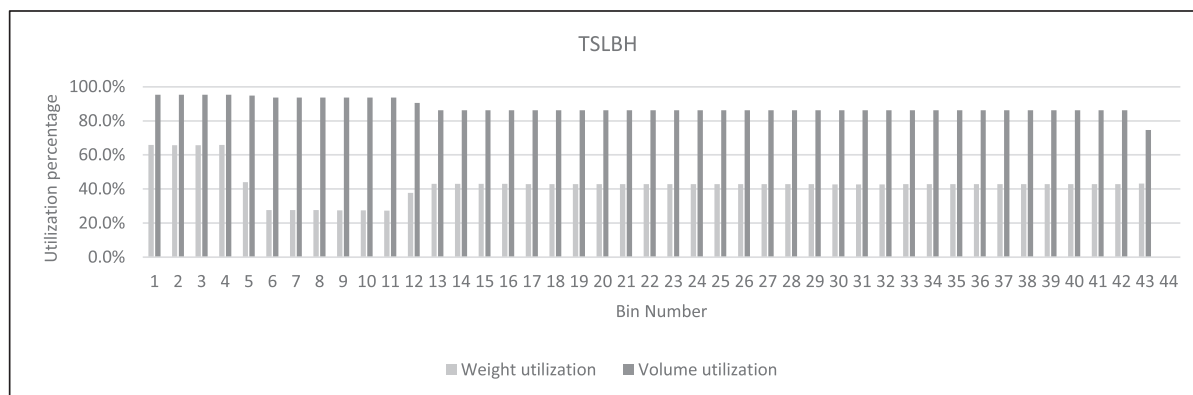


Fig. 5. Load balancing and volume utilization percentages: TSLBH ($n = 10000$, 3 box types).

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